GOVT 6019: Homework 1 — Data, Variables, Notation

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Part A — Concepts

Question 1: Levels of measurement

For each variable below, state its **level of measurement** (nominal, ordinal, or interval-ratio) and whether it is **discrete or continuous**; flag any that are **binary**. Give one sentence of justification each. A couple of these are genuinely debatable, so the justification matters more than the label.

a. `pid3` from `voter2012.csv` (1 Democrat · 2 Republican · 3 Other)
b. `educ` from `voter2012.csv` (1 No HS · … · 6 Post-grad)
c. `birthyr` from `voter2012.csv` (year of birth, 1921–1994)
d. A country's Polity score (-10 = full autocracy … $+10$ = full democracy)
e. The incumbent party's share of the two-party vote in a district (0–100%)
f. Whether a country experienced a military coup in a given year (yes/no)

Question 2: Units of analysis

In 2012, the states with the highest median household incomes — Maryland, New Jersey, Connecticut, Massachusetts — all voted for Obama, while some of the poorest states — Mississippi, West Virginia, Arkansas — went to Romney. A pundit concludes: "rich Americans vote Democratic."

a. What is the unit of analysis in the *evidence*?
b. What is the unit of analysis in the *claim*?
c. Name the error the pundit is at risk of, and explain how the state-level pattern above can be true even if, within every state, richer individuals are *more* likely to vote Republican.
d. What kind of data would you need to evaluate the claim as stated?

Question 3: Operationalizing "voted"

Suppose the concept you care about is **electoral participation**: did a person vote in the November election? Two common operationalizations:

- **(i) Self-report:** a post-election survey asks, "Did you vote in the November election?"
- **(ii) Validated record:** the survey respondent is matched to an official state voter file, which records whether a ballot was cast.

a. Give one strength and one weakness of each operationalization.
b. Self-reports consistently *overstate* turnout. Is that random or systematic measurement error? In which direction does it push a turnout rate computed from survey responses?

c. Suppose you want to describe turnout among voters under 30. Which operationalization would you lean on, and what could still go wrong with it?

Part B — Notation

Question 4: Summation

a. Simplify each expression as far as possible (n, a, b, and the X_i may appear in your answer); evaluate (iii) as a number.

$$\text{(i)} \sum_{i=1}^{n} a \qquad \text{(ii)} \sum_{i=1}^{n}(a + bX_i) \qquad \text{(iii)} \sum_{i=1}^{4}(2i - 1)$$

b. Here are 2012 presidential turnout rates for five states (percent of the voting-eligible population, rounded): Minnesota 76, Wisconsin 73, New Hampshire 71, Ohio 65, Texas 50. Treating these as $X_1, \ldots, X_5$, compute the average $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, then compute each deviation $X_i - \bar{X}$ and the sum of the deviations.

c. Your answer to (b) is not a coincidence. Show, in two lines of algebra, that for *any* n numbers, $\sum_{i=1}^{n}(X_i - \bar{X}) = 0$.

d. True or false: $\left(\sum_{i=1}^{n} X_i\right)^2 = \sum_{i=1}^{n} X_i^2$. If true, prove it; if false, give the smallest counterexample you can.

Part C — Applied: the VOTER data

These questions use the course dataset `voter2012.csv` and its codebook, both on the course site. Work in a script inside your RStudio project, as in lab.

```
library(tidyverse)
voter <- read_csv("data/voter2012.csv")   # adjust to your project's layout
```

Question 5: First contact

a. How many rows and how many columns does the dataset have? What is the unit of observation?
b. Produce a table of counts by `pid3` and `gender` (one function from lab does this in a single line).
c. Using the codebook, which party–gender combination is the most common in the data, and how many respondents does it contain?
d. How many respondents are missing `pid3`?

Question 6: How old is the electorate?

a. The dataset contains no age variable. Why is `birthyr` enough? Create an `age` variable.
b. What are the youngest and oldest ages in the data, and how many respondents have each?
c. In one sentence: what assumption connects `birthyr` to a respondent's exact age, and why is it harmless for describing this dataset?

Question 7 (stretch): Something is off

Spend ten minutes exploring the dataset with `count()`, the codebook open beside you. **Find one thing that looks odd or hazardous** — a value that should not be averaged, a variable whose numbers point the "wrong" way, anything that would trip up a careless analyst. Present the evidence (a count table is enough) and say in a sentence or two what you would want to do about it. There are several right answers; we repair these in Week 2's lab.

If you want more: Gailmard, ch. 1 and §2.1 (measurement, at exactly this level); Gelman et al., *Red State, Blue State, Rich State, Poor State* (Princeton, 2008) on Question 2's paradox; Ansolabehere & Hersh, "Validation: What Big Data Reveal About Survey Misreporting and the Real Electorate," *Political Analysis* 20(4), 2012, on Question 3.